

Orange Money –

B2W-W2B service

Bank Interface Specifications

Version 3.1

Written by	Validated by	Approved by
Bertrand Collomb-Patton Harold Leboulanger		

Date	\$	Modification	Authors
24/02/2015		Small cosmetic modifications (title of documents) to allow a better understanding	Bertrand Collomb-Patton
15/06/2016	1.3, 5	addition of paragraph on status inquiry	Bertrand Collomb-Patton
29/09/2016	6	few additions	Bertrand Collomb-Patton
15/11/2016	6.2	“transaction status inquiry” and “cancel” samples addition	Bertrand Collomb-Patton
03/05/2017	2.4	Clarifications of definitions	Bertrand Collomb-Patton
29/03/2018	2.4	Modification of default alias format	Bertrand collomb-Patton

SUMMARY

1	INTRODUCTION.....	3
1.1	SERVICE'S OVERVIEW	3
1.2	ARCHITECTURE OVERVIEW.....	4
1.3	B2W SERVICE FLOW REMINDER	5
1.4	PURPOSE OF DOCUMENT	5
2	REGISTRATION MODULE.....	6
2.1	SUBSCRIPTION.....	6
2.2	END OF SUBSCRIPTION	6
2.3	SOAP WEB SERVICE DESCRIPTION	7
2.4	DICTIONARY OF EXCHANGED DATA.....	8
2.5	RETURN CODES	9
2.6	SOAP ENVELOPES	10
2.6.1	WSDL FILE.....	10
2.6.2	SOAP ENVELOPES	10
3	IDLE MODE.....	12
3.1	STATIC MODE	12
3.2	IDLE MODULE.....	12
3.2.1	DICTIONARY OF EXCHANGED DATA.....	12
3.2.2	RETURN CODES	12
3.2.3	WSDL FILE.....	12
3.2.4	CHANGEMENT OF STATE.....	13
4	ORANGE STATUS INQUIRY	14
4.1	TRANSACTIONSTATUSINQUIRY	15
4.1.1	DICTIONARY OF EXCHANGED DATA.....	15
4.1.2	SAMPLE	15
4.2	LASTTRANSACTIONS.....	16
4.2.1	DICTIONARY OF EXCHANGED DATA.....	16
4.2.2	SAMPLE	16
4.3	WSDL FILE.....	16
5	END OF DAY FILE	17
5.1	EXCHANGE BETWEEN OM AND THE BANK.....	17
5.2	FORMAT AND CONTENTS	18

1 Introduction

1.1 Service's overview

The service aims at providing Orange Money customers or retailers with means to remotely exchange money between their Orange Money wallet and their bank account.

The service is available through dedicated Orange Money USSD menus for Orange Money customers at any time of day, even during the daily unavailability period of the bank (if any).

Orange Money users will have to previously subscribe to this service through the Orange Money portal and their bank branch ("**Registration**"), before being able, from the Orange Money menu, to ask for money transfers from/to their bank account(s) to/from their Orange Money wallet ("**Transaction**").

Money transfers and service subscription requests have to comply with Orange Money and bank constraints (e.g.: accounts holder & links, threshold, balance ...).

- An Orange Money wallet is able to receive or send money to/from two (or more) bank accounts, provided that the aliases between the Orange Money Account and the two bank accounts are distinct,

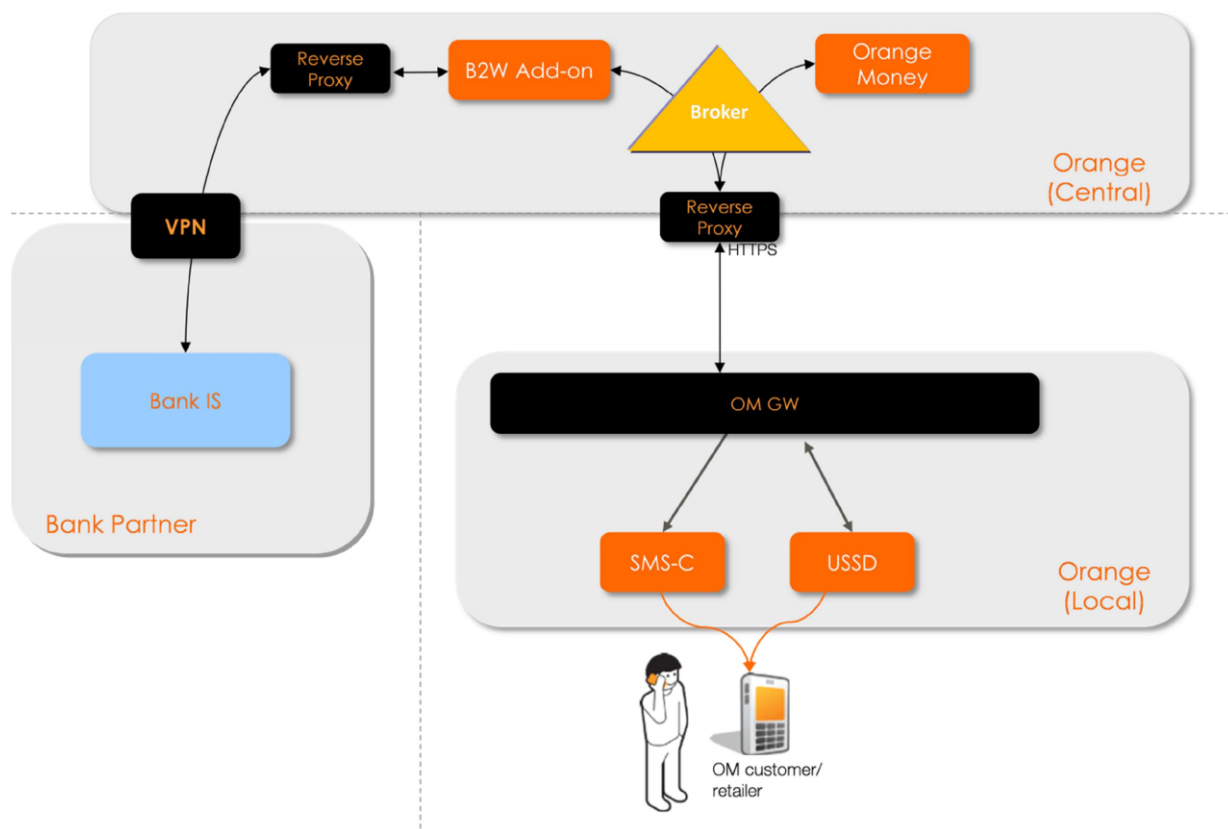
- As well, two distinct OM wallets can send and receive money to/from the same bank account, linked with two distinct aliases,

- For each association (between bank account and OM wallet), the user chooses the authorized operations: OM credit from a bank account (also called "charge"), OM debit to a bank account (also called "discharge"), or both (both is default choice).

On top of these rules, identity of bank and Orange Money accounts' holders has to be checked at service's registration. This check relies on an activation key with a predefined validity period.

This version 3.1 of B2W is associated with the version G4R4C2 of the platform of Orange Money.

1.2 Architecture Overview



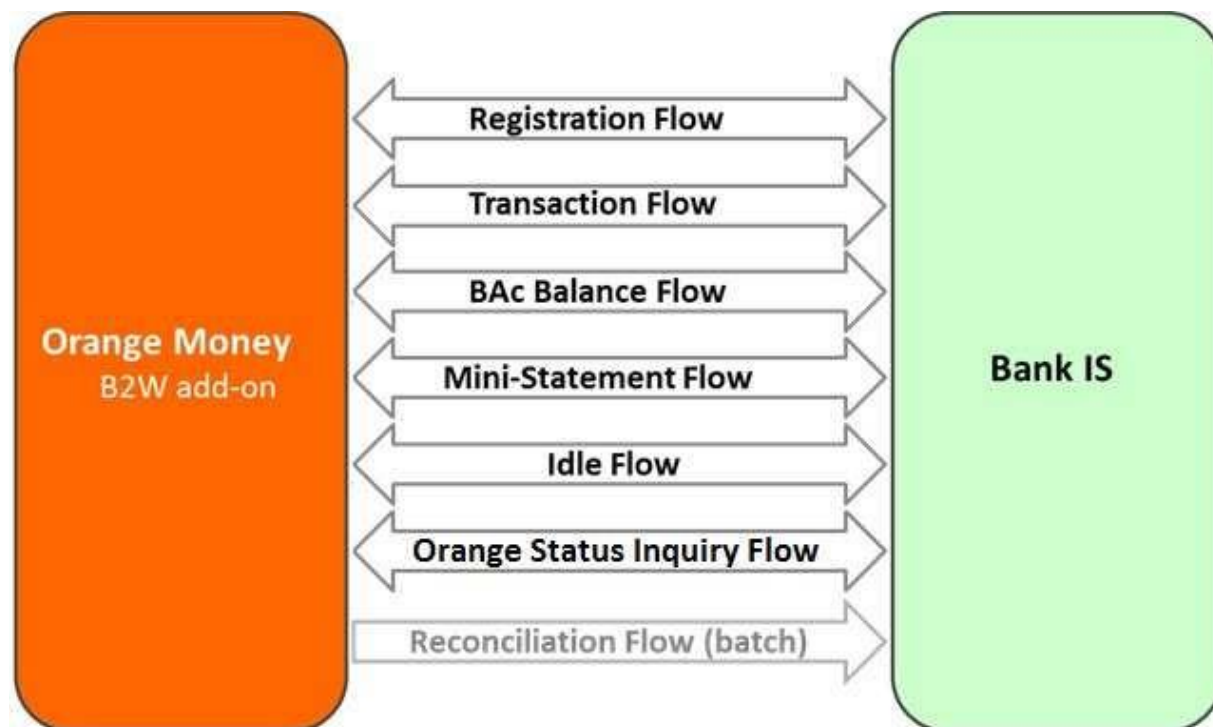
The OM B2W add-on processes user's requests but also provides interfaces between components involved in B2W service:

- Bank partner IS Platform,
- Orange Money Platform,
- Local USSD platform for USSD processing,
- Local SMSC for SMS sending.

Exchanges between Orange and Bank platforms are performed in SOAP:

- OM B2W provides the web services dedicated to registration and idle period management
- Bank Partner provides the web services dedicated to transaction, account balance and mini-statement processing
- Bank Partner and OM B2W add-on respectively requests these web services through their own SOAP client.

1.3 B2W service flow reminder



These flows, except reconciliation flow, are real-time and based on SOAP Web services.

1.4 Purpose of Document

This document is the technical interface specification for the B2W project.

It focuses on the real-time flows:

- Registration flow: subscription to service;
- Transaction flow: 'charge' and 'discharge' transactions;
- Bank account Balance flow : balance enquiry
- Mini-statement flow : mini-statement (last transactions) enquiry
- Idle flow: bank unavailability period's management
- Orange status Inquiry: allows bank partner to know a transaction's state on Orange's side

Reconciliation flow, relying on daily file transmission, is also described.

Definition of flows relies on B2W service's use cases as defined in document:

OM-B2W Service description - V3.1.docx.

2 Registration module

2.1 Subscription

This version 3.1 of B2W service proposes to banks 2 modes of users' subscription:

- **One with a comparison of KYC presented by users at the bank and those presented at Orange at the OM's subscription,**
- **One without this comparison.**

Subscription takes place into 2 stages, consisting in:

- First stage:
 - ⇒ The Orange Money user requests Orange Money add-on, through a dedicated USSD menu, for subscribing the B2W service for a not yet registered bank account.
 - ⇒ If the request can be successfully processed by Orange Money add-on (identity check OK, OM wallet OK, number of registered accounts...), he would get in return an activation key.
- A second stage:
 - ⇒ The user requests a bank agent to subscribe to the B2W service for a not yet registered bank account, he holds in the bank.
 - ⇒ Once the bank agent has the activation key, he submits the KYC request to bank IS.
 - ⇒ **Bank IS processes the KYC request and transmits it to OM B2W add-on.**
 - ⇒ **B2W add-on processes the received request and sends to bank IS the KYC of the user in return.**
 - ⇒ **the bank agent can compare KYC sent by Orange with those given by the user.**
 - ⇒ Once the bank agent has checked user's identity, he submits the subscription request to bank IS.
 - ⇒ Bank IS processes the request and then transmits it to OM B2W add-on.
 - ⇒ B2W add-on processes the received registration request and sends to bank IS the appropriate response code in return.

These steps involve bank IS & OM add-on interface.

2.2 End of Subscription

Un-subscription may be issue by:

- User: requesting a bank agent to terminate the subscription;
- Bank: for private reasons beyond the control of user;
- Orange: for private reasons beyond the control of user too.

The 2 first cancellation kinds involve exchanges through the bank Is & OM add-on interface. The last case involves batch exchanges inside daily file (FQE).

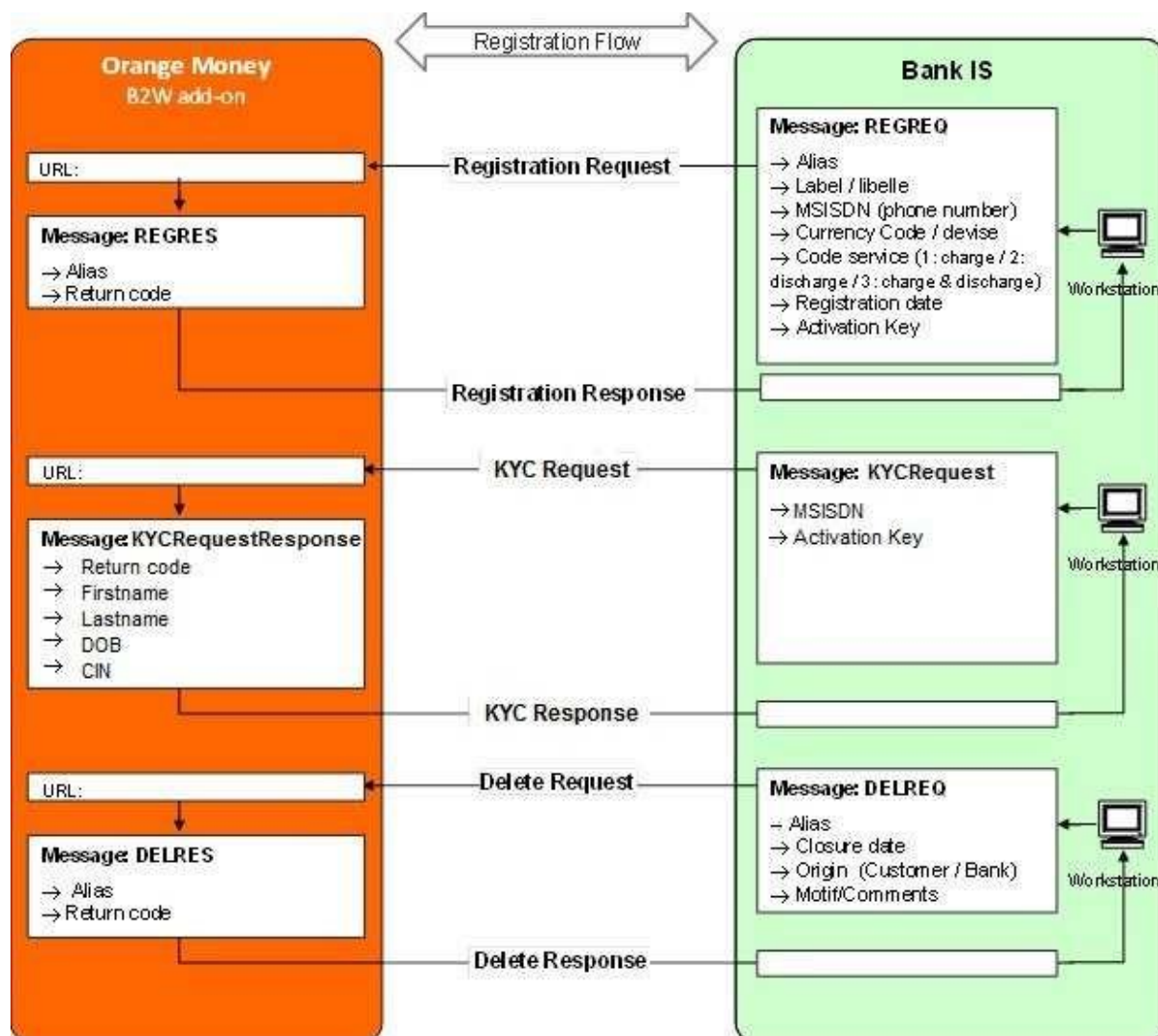
2.3 SOAP Web Service description

Two kinds of messages dealing with registration process are available: subscription and closure requests.

These SOAP messages are defined in a WSDL file to considered bank partner, which serves as recommendation and describes:

- The used protocol : SOAP/RPC
- The message format required to communicate with the service,
- The methods the client can invoke: ombRequest(), ombClose(), KycRequest()
- The service localisation

(Here the URL of the targeted service is the development platform. Addresses of service testing platform, pre-prod and prod will be specified later).



2.4 Dictionary of exchanged data

Alias format:

Default Account Alias is 22 Alpha-numeric characters:

Default Account Alias = Bank BIC Code (8 digits) + Customer ID (14 numerical digits)

However, banks can prefer to use different format.

Example: BICBANKX25486357988514

	Field	length	Format	SOAP	Example	Description
1	alias	22 (bank choice)	Alpha	String	BICBANKX25486332888514	This field contains the alias allowing to identify the link between bank account and OM account
2	msisdn	7, 10 (ex)	Num	String		OM account number : National MSISDN format (without local prefix) ML : 7 digits OCI, SN, MG
3	code_service	1	Num	Int	3	This field corresponds to the authorized operation types : • Charge: 1 • Discharge: 2 • All : 3
4	label	0 to 25	Alpha	String	Personal account	This field corresponds to the label displayed in the customer USSD menu. Special characters are disabled
5	active_date & close_date	19	Timestamp ISO 8601	DateTime	2017-01-31 16:05:22	This field corresponds to the current date and time. Date format is YYYY-MM-DD with • YYYY = Year • MM = Month • DD = Day Time format is HH:MM:SS with • HH = Hour • MM = Minute • SS = Second Date and time are UTC ones
6	devise	3	Num	String	952/969/...	ISO num currency • 952=XOF • 969=MGA
7	Key	8	Alpha	String	D67HM235	Activation key. OM account type prefix • D = distributor / retailer • C = customer
8	motif	0 to 200	Alpha	String	Closure upon customer request	Closure comments. Free Optional field.
9	origin	4 / 6	Alpha	String	Client	Request initiator. Accepted values : Client Bank Orange
10	return_code	3	Num	Int	200	Cf. return codes

Only WSDL-specific file is recommended to consume the dedicated web service.

2.5 Return codes

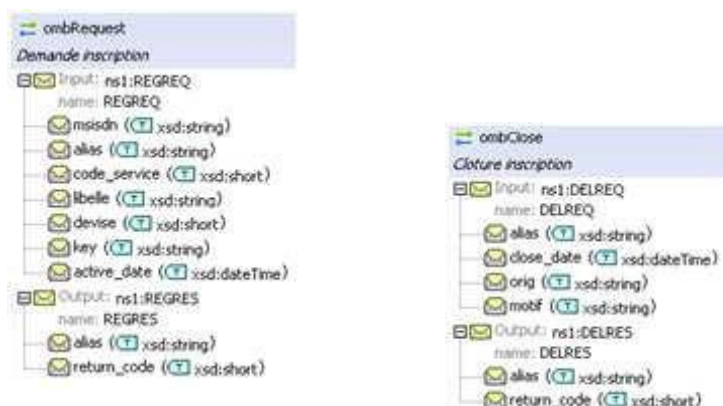
Return code	Description	REGRES	DELRES
200	OK: validation	X	X
302	KO: invalid request	X	X
303	KO: Wrong MSISDN format	X	
304	KO: Unknown alias	X	X
306	Invalid Currency	X	
307	Invalid service code	X	
601	KO: Alias already exists	X	
602	KO: Unknown request or timeout	X	
603	KO: Invalid activation key	X	
604	KO: Platform unavailable	X	X

2.6 SOAP Envelopes

The WSDL is named Register.wsdl.

It propose 1 subscription, 1 unsubscription and 1 getKYC methods.

WSDL diagrams below define subscription and unsubscription SOAP requests, gathering by message exchanged data previously described.



2.6.1 WSDL file

WSDL file name is **register-kyc.wsdl**

3 Idle mode

The bank can define periods during which ones it's not available.

During these periods, B2W add-on module stocks requests, and transmit them at the end of unavailability periods.

3.1 Static Mode

The bank can define one daily period during which one it's not available.

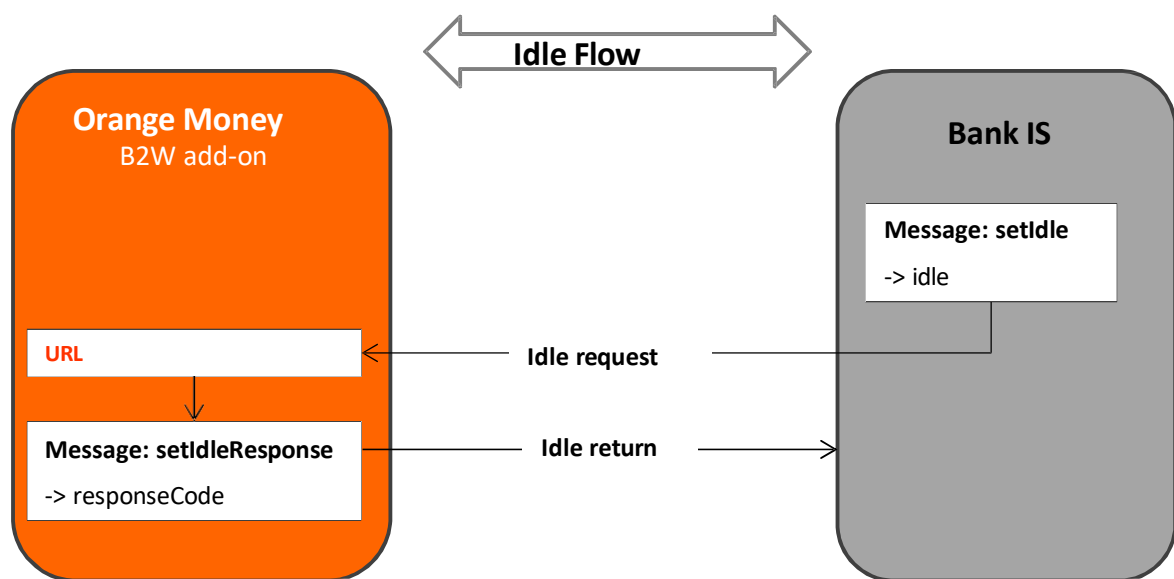
Hours of unavailability are fixed in configuration file.

e.g.: from 23:00 to 01:00.

3.2 Idle module

The Idle Module is defined to dynamically manage bank daily unavailability period.

It relies on a single message allowing the bank IS to advise the OM B2W add-on of start & end of unavailability period: 'setIdle'.



3.2.1 Dictionary of exchanged data

The bank status is the single data expected for being exchanged.

Field	I/O*	Length	Format	Description
idle	O	4/5	Bool	"true" or "false" (setIdle) In targeted case, "true" means bank is unavailable
responseCode	I	4/5		"true" in case of success, "false" otherwise (setIdleReponse)

3.2.2 Return codes

No return codes are defined for "idle" messages.

Returns "true" if successful, "false" otherwise. On error, the state of the platform remains unchanged.

3.2.3 WSDL file

The WSDL file is Idle.wsdl.

3.2.4 Chagement of state

Allow the external partner (Bank) to disable the above data transmission (for transactions) during the unavailability period.

Request

```
<soapenv:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:om="http://om.banking.com">
  <soapenv:Header/>
  <soapenv:Body>
    <om:setIdle soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
      <idle xsi:type="xsd:string">false</idle>
    </om:setIdle>
  </soapenv:Body>
</soapenv:Envelope>
```

- Response

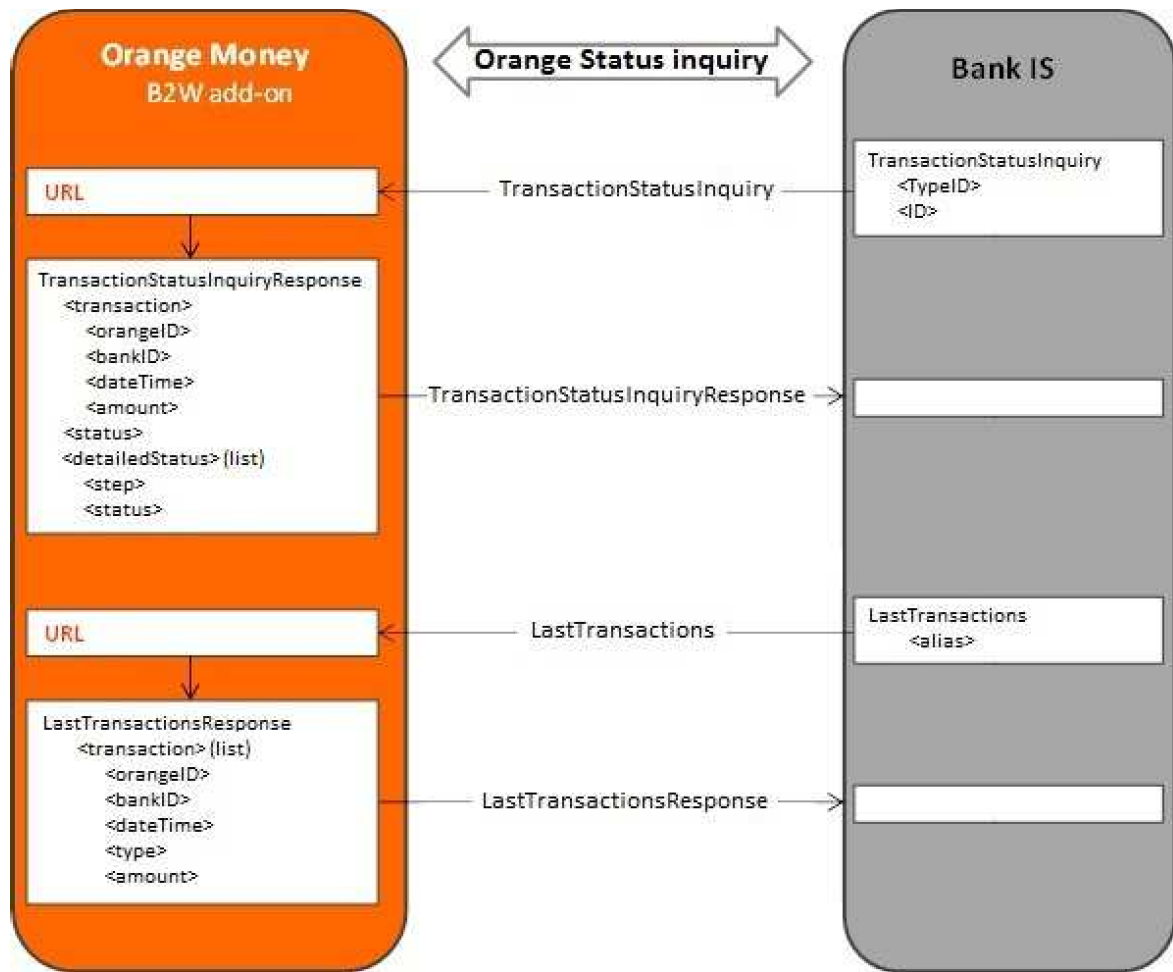
Returns "true" if successful, "false" otherwise. On error, the state of the platform remains unchanged.

```
<soapenv:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:om="http://om.banking.com">
  <soapenv:Header/>
  <soapenv:Body>
    <om:setIdleResponse soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">
      <responseCode xsi:type="xsd:boolean">true</responseCode>
    </om:setIdleResponse>
  </soapenv:Body>
</soapenv:Envelope>
```

4 Orange Status Inquiry

The bank can request status of operations on Orange side.

The Module is defined to inform the bank partner of status of transaction.



4.1 TransactionStatusInquiry

This method allows the bank partner to require a status of an transaction by entering either a bank transaction identifier, or an Orange transaction id.

4.1.1 Dictionary of exchanged data

Field	SOAP	Sample	Description
TypeID	String	BANK_ID	Defined if the Id is an Orange or bank one: - BANK_ID - ORANGE_ID
ID	String	OR1236456 MBDCH.12687987.132	One identifier (bank or Orange) of the wanted transaction.
orangeID	String	MBDCH.12687987.132	Orange identifier of the transaction
bankID	String	OR1236456	Bank identifier of the transaction
dateTime	String	2015-03-13T13:37:45.502+01:00	Date of the transaction
amount	String	1500.25	Amount of the transaction
status	String	KO	Global status of the transaction: - OK - ORANGE_RECONCILIATION - BANK_RECONCILIATION - KO
List of detailed status			Description of the successive steps, with theirs status
step	String	BANK_CREDIT	Identifier of the successive steps: - ORANGE_CREDIT - ORANGE_DEBIT - BANK_CREDIT - BANK_DEBIT - BANK_STATUS_INQUIRY
status	String	OK	detailed status of the step of the transaction - OK - TIMEOUT - KO

4.2 LastTransactions

This method allows the bank partner to require information on last transactions of an user, in order to, then, require detailed information (TransactionStatusInquiry) on one of these last transactions.

4.2.1 Dictionary of exchanged data

Field	SOAP	Sample	Description
alias	String	ECOCMLBA120123000000	Alias of the user
List of last transactions			Detailed description of each transaction
orangeID	String	MBDCH.12687987.132	Orange identifier of the transaction
bankID	String	OR1236456	Bank identifier of the transaction
dateTime	String	2015-03-13T13:37:45.502+01:00	Date of the transaction
type	String	WALLET2BANK	Type of the transaction: - BANK2WALLET - WALLET2BANK
amount	String	1500.25	Amount of the transaction

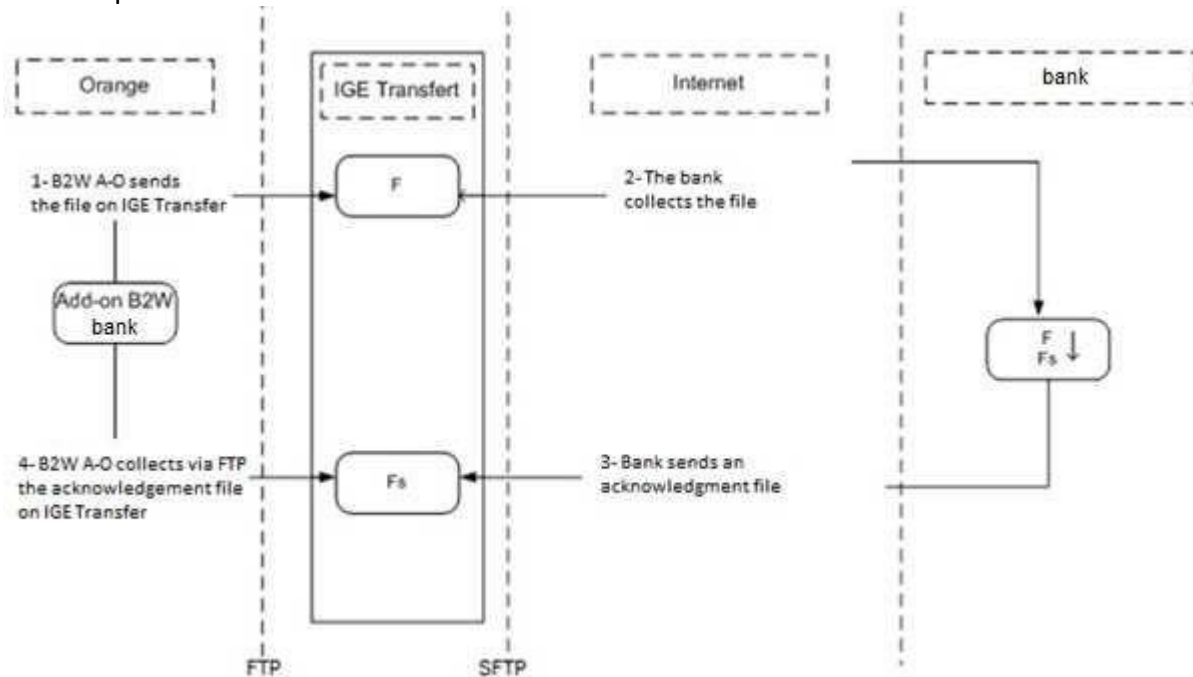
4.3 WSDL file

The WSDL file is statusInquiry.wsdl.

5 End of Day file

5.1 exchange between OM and the bank

This file is transmitted to the bank partner each end-of-day (only business days: tbc) using secured protocol SFTP.



- **Periodicity:** defined by a calendar with holidays.
Example: 6/7 from Monday to Saturday, except holidays.
Days without FQE are added in order to appear in the following FQE.
- **File name:** B2W_BANKNAME_COUNTRY_FQE_YYYYMMJJHHmm.txt
- **Hours of transfer:** generation and transfer hour of FQE is configurable.
- **records management:**
The file contains no doublon.
 - One raw per alias for subscription,
 - One raw per alias for unsubscription,
 - One raw per processing code and final status,
 - Empty file: 1 header and 1 end only.
- **Retry management:**
 - If the file transfer on IGE platform fails, Orange retry 3 times (number configurable).
Interval between retries is configurable.
 - Then, Orange tries the following day, once.
 - If there is still a fail, we're doing a manual file exchange.

- FQE's acknowledgment of the bank for Orange

- Return code is written in the file

ACK_B2W_BANKNAME_COUNTRY_FQE_YYYYMMJJHHmm.txt.

CODE	Description
200	File status okay
300	Data FQE corrupted
404	FQE Not found
500	Can't open data connection.

- A end code (\n\r or le Hexa code 0D 0A) is added after the error code.
- One hour (configurable time) after having sending the FQE, Orange attempts to get the acknowledgment on IGE Transfer PF.
- If it fails, 3 (configurable) retries are done. (Interval is configurable.)
- Without this file, Orange considers that the FQE is not get by the bank. In this case, a manual action is done.

5.2 Format and contents

Format: XML file

It lists all operations performed from the last EoD file transmission:

- Subscriptions
- Un-subscriptions (on subscriber or bank initiative, and on Orange initiative)
- Transactions: recharge, discharge, cancellations of transaction, bank balance, mini-statement

It consists of:

- 1 header
- m recordings of partner requests
- n recordings of aliases history (subscription / unsubscription)

Samples

```
<eod>
  <header>
    <datetime>2016-12-08T07:54:37+00:00</datetime>
    <partnerRequestsCount>3</partnerRequestsCount>
    <aliasesHistoryCount>3</aliasesHistoryCount>
  </header>
  <partnerRequests>
    <partnerRequest><type>balance</type><alias>BICBICX00000000001100</alias><amount></request
      Code>OR1412054A9ZG2</requestCode><responseCode>000</responseCode><referenceNumber/
      ><requestDate>2016-12-05T07:56:00+00:00</requestDate></partnerRequest>
    <partnerRequest><type>statement</type><alias>BICBICX00000000002100</alias><amount></requ
      estCode>OR141205AEJ7GH</requestCode><responseCode>000</responseCode><referenceNumb
      er></requestDate>2016-12-05T07:56:07+00:00</requestDate></partnerRequest>
    <partnerRequest><type>bank2wallet</type><b2wid>MBRCH.20161114181747.983</b2wid><alias>BI
      CBICX000000000013000</alias><amount>1000.00</amount><requestCode>OR161114261IGN</
      requestCode><responseCode>000</responseCode><referenceNumber>MOCK29716</referenceN
      umber><requestDate>2016-11-14T18:17:48+00:00</requestDate></partnerRequest>
    <partnerRequest><type>transaction-status-
      inquiry</type><b2wid>MBRCH.20161114181747.983</b2wid><alias>BICBICX000000000013000</
      alias><amount>1000.00</amount><requestCode>OR161114BSPFG5</requestCode><responseCod
      e></responseCode><referenceNumber></referenceNumber><requestDate>2016-11-
      14T18:17:59+00:00</requestDate></partnerRequest>
    <partnerRequest><type>cancel-
      b2w</type><b2wid>MBRCH.20161114181747.983</b2wid><alias>BICBICX000000000013000</ali
      as><amount>1000.00</amount><requestCode>OR1611141BFGJ2</requestCode><responseCode>
      000</responseCode><referenceNumber>MOCK5722</referenceNumber><requestDate>2016-11-
      14T18:08:07+00:00</requestDate></partnerRequest>
  </partnerRequests>
  <aliasesHistory>
    <aliasHistory><type>register</type><alias>BICBICX7509464464583</alias><responseCode>603</re
      sponseCode><requestDate>2016-12-05T07:54:59+00:00</requestDate><bankDate>2016-12-
      05T07:53:27+00:00</bankDate></aliasHistory>
    <aliasHistory><type>register</type><alias>BICBICX00000000002200</alias><responseCode>200</re
      sponseCode><requestDate>2016-12-05T07:55:24+00:00</requestDate><bankDate>2016-12-
      05T07:55:23+00:00</bankDate></aliasHistory>
    <aliasHistory><type>close</type><alias>ECOCMLBA00000000000019</alias><responseCode>200</re
      sponseCode><requestDate>2016-02-09T16:08:50+00:00</requestDate><bankDate>2016-02-
      09T16:08:50+00:00</bankDate><closeOrigin>Client</closeOrigin><closeReason>test</closeReaso
      n></aliasHistory>
  </aliasesHistory>
</eod>
```

</type> list

- for </partnerRequest>:
wallet2bank, bank2wallet, cancel, cancel-w2b, cancel-b2w, balance, statement, transaction-status-inquiry.
- for </aliasHistory>:
register, close.